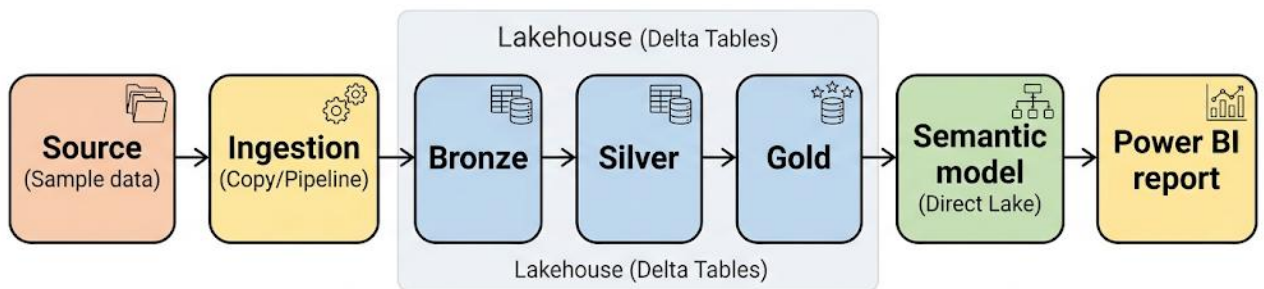


NYC Taxi Data Engineering Capstone Project

1. Project Overview

This project demonstrates an end-to-end Data Engineering pipeline using Microsoft Fabric. The NYC Green Taxi dataset was ingested into Bronze, transformed into Silver, modeled into Gold in star schema tables, and visualized using Power BI.

2. Architecture



3. Tech Stack

1. Microsoft Fabric
2. Lakehouse
3. Python/SQL/Pyspark
4. Power BI
5. Direct Lake Semantic Model

4. Bronze Layer

1. Raw NYC Taxi data ingested into Lakehouse
2. No transformations applied
3. Initial profiling completed:
4. Row count
5. Null checks
6. Duplicate checks

5. Silver Layer

Performed data cleaning and standardization:

- Fixed data types
- Removed duplicates
- Handled null values
- Standardized column names
- Created Date Dimension
- Applied validation rules

6. Gold Layer – Data Dictionary

1. Table Name: ashish_gold_fact_trips

Description : This fact table stores detailed NYC Green Taxi trip transaction records.

Grain: One row per taxi trip transaction.

Key Columns:

Column Name	Explanation
p_date_key	Foreign key linked to pickup date in Date Dimension table.
do_date_key	Foreign key linked to dropoff date in Date Dimension table.
vendor_id	Identifies the taxi service vendor associated with the trip.
pickup_location_id	Represents the pickup zone/location ID.
dropoff_location_id	Represents the dropoff zone/location ID.
pickup_dropoff_pair	Combined pickup and dropoff route pair used for route analysis.
passenger_count	Number of passengers in the trip.
trip_distance	Distance travelled during the trip in miles.
fare_amount	Base fare amount charged for the trip.
tip_amount	Tip amount paid by the passenger.
tolls_amount	Toll charges applied during the trip.
total_amount	Total transaction amount including all charges.

2. Table Name: ashish_gold_dim_date

Description : This dimension table stores calendar and date-related attributes.

Key Columns:

Column Name	Explanation
date_key	Primary key generated in YYYYMMDD format for relationship mapping.
date	Complete calendar date value.
year	Year extracted from the date column.
quarter	Quarter of the year (1–4).
month	Numeric month value (1–12).
month_name	Full month name used for reporting and visualizations.
day	Day of the month.
day_name	Name of the weekday.
weekday	Numeric representation of weekday where Monday = 0 and Sunday = 6.

3. Table Name: ashish_gold_dim_pickup_location

Description : This dimension table stores unique pickup location information used for pickup zone analysis.

Key Columns:

Column Name	Explanation
pickup_location_id	Unique identifier representing pickup zones.
pickup_location_name	Generated descriptive location label for visualization and reporting.

4. Table Name: ashish_gold_dim_dropoff_location

Description : This dimension table stores unique dropoff location information used for dropoff zone analysis.

Key Columns:

Column Name	Explanation
dropoff_location_id	Unique identifier representing dropoff zones.
dropoff_location_name	Generated descriptive location label for visualization and reporting.

6. Data Quality Rules

Rule Type	Column(s)	Rule Applied	Action Taken
Completeness	puLocationId, doLocationId	Must not be null	Null rows removed
Uniqueness	pickup/drop datetime + vendor + locations + fare	Duplicate trips removed	Business-key deduplication
Validity	passengerCount	Must be ≥ 1	Invalid rows removed
Validity	Monetary columns	Negative values converted using abs()	Standardized numeric values
Datetime Validation	pickup/drop datetime	Dates must not exceed current date	Future records removed
Logical Validation	pickup/drop datetime	Dropoff > Pickup	Invalid trips removed

7. Validation Checks

1. Validation Check	Result
2. Silver vs Gold row counts	Matched
3. Duplicate dimension keys	No duplicates found
4. Fact table foreign key coverage	Valid relationships established
5. Semantic model relationships	Successfully validated
6. Measures responding to slicers	Passed

8. Semantic Model

A Direct Lake Semantic Model was created using the Gold layer tables stored in Microsoft Fabric Lakehouse. The semantic model was designed using a star schema approach consisting of one fact table and multiple dimension tables.

The fact table `ashish_gold_fact_trips` was connected with the following dimension tables:

1. `ashish_gold_dim_date`
2. `ashish_gold_dim_pickup_location`
3. `ashish_gold_dim_dropoff_location`

Relationships were configured between fact and dimension tables using primary and foreign keys to support efficient analytical reporting and filtering.

Separate pickup and dropoff date keys were maintained in the fact table to support role-playing date relationships within the semantic model.

Multiple DAX measures and KPIs were created inside a dedicated Measure table for business analysis and reporting purposes.

Measures Created

1. Total Trips
2. Total Fare Amount
3. Total Distance
4. Total Passengers
5. Trips on Working Days
6. Trips on Weekend Days
7. Total Fare on Working Days
8. Total Fare on Weekend Days
9. Total passengers on Working Days
10. Total passengers on Weekend Days
11. Rate per Mile

The semantic model was later connected to Power BI report pages for interactive dashboard creation and business insights generation.

9. Power BI Report

The Power BI report was created using the Direct Lake Semantic Model developed in Microsoft Fabric. The dashboard was designed with interactive visuals, slicers, KPIs, and analytical charts to analyze NYC Green Taxi trip patterns and business insights.

Page 1 – Summary Dashboard

The first report page focuses on high-level trip and fare analysis using KPI cards and summary visuals.

Visuals Included

1. Total Trips KPI
2. Total Fare Amount KPI
3. Working Day vs Weekend Trip Analysis
4. Top Pickup Locations Bar Chart
5. Top Dropoff Locations Bar Chart
6. Date Range Slicer
7. Day-based Filter Slicer

Page 2 – Detailed Analysis Dashboard

The second report page focuses on route-level and trend-based analysis using detailed visuals and matrix reporting.

Visuals Included

1. Pickup to Dropoff Pair Matrix
2. Underperforming Pickup Location Analysis
3. Distance, Total Fare amount and Revenue KPI Cards
4. Working Day vs Weekend total Fare Analysis
5. Working Day vs Weekend total passengers Analysis

10. Steps to Reproduce

1. Upload the NYC Green Taxi parquet dataset into the Microsoft Fabric Lakehouse.
2. Run the Bronze layer notebook to ingest the raw dataset and perform initial profiling.

3. Run the Silver layer notebook to clean and transform the data by handling null values, duplicates, datatypes, and validation rules.
4. Save the transformed dataset as: **ashish_silver_trips**
5. Run the Gold layer notebook to create:
 1. ashish_gold_fact_trips
 2. ashish_gold_dim_date
 3. ashish_gold_dim_pickup_location
 4. ashish_gold_dim_dropoff_location
6. Build and refresh the Direct Lake Semantic Model in Microsoft Fabric.
7. Create DAX measures and Power BI report visuals using the semantic model.
8. Refresh the report and validate KPIs, relationships, and dashboard visuals.

11. Future Improvements

1. Integrate the official NYC Taxi Zone lookup dataset for accurate zone names and geographical mapping.
2. Create a separate DimTime table for hour-level and peak-hour trip analysis.
3. Implement incremental refresh to optimize large dataset processing.
4. Add Data Factory pipelines for automated ingestion and orchestration.
5. Include advanced DAX calculations such as MoM and YoY trend analysis.

12. Daily Progress Log

1. 11/5/2026:

1. what you did?

I read and understood the shared capstone project guide document.

2. what issues you found?

The Project aims to complete within 2 weeks and should only use Microsoft Fabric facility as per the company.

3. what you will do next?

Create a new account to access free trial in Microsoft Fabric.

2. 12/5/2026:

1. what you did?

Created a free account in Microsoft Fabric.

2. what issues you found?

Accessing free trial in Microsoft Fabric requires too much validation.

3. what you will do next?

Ingest and understand NYC taxi data.

3. 13/5/2026:

1. what you did?

I ingested the NYC Green Taxi parquet dataset into Microsoft Fabric notebook environment and performed initial data profiling using Pandas. I analyzed schema, row count, null values, duplicate records and descriptive statistics to understand the dataset structure.

2. what issues you found?

The dataset contained multiple data quality issues such as null values, duplicate records, negative fare values, invalid passenger counts and unrealistic datetime values.

3. what you will do next?

Start building the Silver layer by performing data cleaning, datatype validation and business rule filtering.

4. 14/5/2026:

1. what you did?

I started the Silver layer transformation process by creating a working copy of the Bronze dataframe and performed duplicate analysis using business keys such as pickup datetime, dropoff datetime, location IDs and fare amount.

2. what issues you found?

Some columns contained mixed datatypes and object values instead of numeric datatypes.

3. what you will do next?

Handle null values, standardize datatypes and apply business validation rules for numeric and datetime columns.

5. 15/5/2026:

1. what you did?

Handled high null value columns, converted required columns into proper datatypes and applied multiple business rules such as validating positive trip values, valid passenger count, non-null location IDs and valid pickup/dropoff datetime conditions.

2. what issues you found?

Certain columns contained corrupted or unrealistic values including future dates and negative transactional amounts which required additional validation and cleaning logic.

3. what you will do next?

Standardize column naming conventions, create the Silver table and start designing Gold layer dimension tables.

6. 18/5/2026:

1. what you did?

I created the Gold layer star schema tables using the cleaned Silver dataset. I built the **ashish_gold_fact_trips**, **ashish_gold_dim_date**, **ashish_gold_dim_pickup_location**, and **ashish_gold_dim_dropoff_location** tables and defined the grain as one row per taxi trip transaction.

2. what issues you found?

While designing the Gold layer, I faced challenges related to date key generation and maintaining proper relationships between fact and dimension tables.

3. what you will do next?

Create the Semantic Model in Microsoft Fabric and establish relationships between fact and dimension tables.

7. 19/5/2026:

1. what you did?

I created the Direct Lake Semantic Model using the Gold layer tables and established relationships between the fact table and dimensions. I configured active and inactive date relationships and validated the star schema structure.

2. what issues you found?

I encountered relationship conflicts while connecting multiple date keys with the same Date dimension table and resolved them using active and inactive relationships.

3. what you will do next?

Create DAX measures and KPIs for trip analysis, fare analysis, and pickup/dropoff location insights.

8. 20/5/2026:

1. what you did?

I created multiple DAX measures including total trips, total fare amount, total passengers, trips on working days, trips on weekend days, fare on working days, fare on weekend days, passengers on working days, passengers on weekend days total distance, and rate per mile.

I also created a dedicated measure table to organize all KPI calculations.

2. what issues you found?

Some DAX functions returned table expressions instead of scalar values, causing errors while using card visuals and measures in the semantic model.

3. what you will do next?

Build Power BI report pages with KPI cards, charts, slicers, matrices, and detailed analytical visuals.

9. 21/5/2026:

1. what you did?

I designed the Power BI report with multiple visuals including KPI cards, bar charts, pie charts, line charts, slicers, and matrix visuals. I created separate summary and detailed analysis pages focusing on pickup locations, dropoff locations, fare trends, and pickup-dropoff route analysis.

2. what issues you found?

Some visuals initially displayed incorrect aggregations because of relationship direction and measure configuration issues inside the semantic model.

3. what you will do next?

Perform validation checks, finalize dashboard formatting

10. 22/5/2026:

1. what you did?

I completed validation checks between Silver and Gold layers, verified semantic model relationships, and finalized the README documentation including architecture flow, data dictionary, quality rules, validation checks, semantic model explanation, and Power BI report screenshots.

2. what issues you found?

Maintaining consistency between notebook transformations, semantic model relationships, and dashboard calculations required additional reconciliation and validation.

13. Author

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